

Mahesh Gaire || Java Software Engineer

Email: Mahesh.gaire07@gmail.com

Phone : (703) 543-0110 || [Linkedin](#)

Professional Experience:

Full Stack Java Developer with 6+ years of experience delivering scalable enterprise applications using **Java** and **Spring Boot** in cloud-native, microservices-based environments. Focused on building secure, testable, and modular backends while also developing responsive frontend interfaces using **Angular** and **TypeScript**. Skilled in designing RESTful architectures and deploying distributed systems on **AWS**, with hands-on use of services like **ECS**, **Lambda**, **RDS**, and **DynamoDB** to support high-availability solutions. Actively contribute to **Agile/Scrum** teams, participating in sprint planning, retrospectives, and technical discussions to align goals and resolve blockers efficiently. Strong advocate of clean code and **CI/CD pipelines**, using tools like **Jenkins** and **Git** to support continuous integration and reliable deployments. Take ownership of deliverables and collaborate cross-functionally with QA, DevOps, and product teams to drive business outcomes. Known for clear communication, a proactive mindset, and consistent focus on quality, maintainability, and **system performance**.

Technical Skills:

Languages	Java 8/11/17/21, SQL, PL/SQL, TypeScript, JavaScript, Python
Frameworks & Technologies	Spring (Core, MVC, Boot, Security, DAO, Batch, Cloud, WebFlux), Hibernate, JPA, Microservices, Swagger / OpenAPI, Cucumber, JUnit, TestNG, Mockito, RESTful APIs, SOAP
Web Technologies	Angular 10/14/16/19, HTML5, CSS3, Bootstrap, JQuery, AJAX, JSON, XML
Cloud Computing	AWS (EC2, S3, EBS, RDS, IAM, CloudWatch, DynamoDB, ECS, Lambda), Kubernetes,
DBMS	Oracle, PostgreSQL, MySQL, MongoDB
Web/Application Servers	Apache Tomcat
Messaging & Streaming:	Kafka, Kafka Streams, JMS, RabbitMQ
DevOps & CI/CD Tools	Jenkins, Docker, Git, OpenTelemetry, Prometheus, Grafana, Splunk, Bitbucket, Maven, SonarQube, Terraform
Testing Tools	JUnit, TestNG, Postman, Swagger UI
IDE	Eclipse, IntelliJ
Built Tools	Maven
Security & Authentication	JWT, OAuth2
Operating Systems	UNIX, Linux, Windows
Version Controls	Git, SVN

Education:

Masters in Computer Science
University of the Cumberland's

Bachelors of Science in Computer Science
Kent State University

Professional Experience:

Client: E*TRADE by Morgan Stanley

Jul 2022 to Present

Role: Senior Software Engineer

Project: Market Data Aggregator

Developed and enhanced backend microservices responsible for **aggregating**, normalizing, and distributing real-time market data from multiple financial vendors. Delivered high-throughput, fault-tolerant systems leveraging **Java**, **Spring Boot**, **Spring WebFlux**, and **Kafka**, with front-end dashboards in **Angular** to visualize latency, symbol coverage, and vendor performance.

Responsibilities:

- Upgraded the core test automation framework to **Java 17** and enhanced it with **Spring Boot** so that backend components could expose reusable testing services via **REST APIs**.
- Developed **Angular 16** components using **TypeScript** to visualize real-time test execution metrics and API statuses consumed from Spring Boot services.
- Upgraded backend services from **Java 17 to 21** by using virtual threads and structured concurrency for optimized performance.
- Developed feature flags using **TypeScript** in **Angular** modules so that QA users could enable or disable specific test logic without code changes.
- Utilized **TypeScript** and **RxJS** to build reactive UI components that subscribe to backend observables exposed by Spring Boot services.
- Implemented **Angular state** management using **NgRx** to maintain consistent test execution state and reactive data flow across components.
- Explored **Angular 19** features by using standalone components and enhanced reactivity for modular UI improvements.
- Implemented **Kafka** producers and consumers with schema validation to manage asynchronous transactions for message driven processing between distributed microservices.
- Configured **Kafka** topic monitoring through offset tracking and retry templates to achieve fault recovery for delayed or failed message deliveries.
- Developed and optimized **MySQL**-backed Spring Boot microservices using **JPA** repositories, transactional queries, and schema mappings to support high-performance data retrieval.
- Configured **Grafana** dashboards with Prometheus metrics and **OpenTelemetry** traces to visualize service latency, throughput, and error rates.
- Implemented **Kafka-based** event-driven microservices using **Spring Kafka**, schema validation, and retry mechanisms to achieve fault-tolerant message processing across distributed systems.
- Built an internal web service using **Spring WebFlux** to expose high-concurrency endpoints for test utilities like PICT generation and Jira integration without timeouts.
- Created a non-persistent data layer using an in-memory key-value store backed by **ConcurrentHashMap** to eliminate external DB dependencies during test runs.
- Containerized multiple backend services using **Docker** for deployment into **AWS ECS**, implementing environment-specific overrides to simplify configuration management across SIT and UAT environments.
- Monitored system performance and test behavior using **Splunk** by pushing structured JSON logs from backend services, including memory and thread statistics.

- Developed, tested, and documented **RESTful** APIs using **Postman** for validation and **Swagger** (OpenAPI) for clear, standardized API documentation.
- Utilized **AWS IAM** roles and policies to control access to ECS services, S3 buckets used for test artifacts, and **CloudWatch** logging groups.
- Implemented **AWS API Gateway** with **OAuth2/JWT** authentication and rate-limiting to securely route traffic to Spring Boot microservices running on AWS.
- Used AI-assisted tools like **GitHub Copilot** to enhance code efficiency and integrated **machine learning APIs** into applications for feature development.
- Enhanced security by applying **RBAC** policies to control user-level access to **REST APIs** and internal test utilities.
- Integrated **Amazon S3** with the automation framework to dynamically fetch and cache test data, configuration files, and reusable input sets.
- Created **AWS Lambda** functions using **Python** triggered by **S3** events to process incoming test artifacts and push metadata into **DynamoDB** for quick lookups.
- Integrated **MongoDB** as a **NoSQL** data store for caching, logging, and schema-flexible test configurations.
- Refactored **Cucumber-based** automation by integrating **TestNG**, **Mockito**, and structured logging so that test coverage and error diagnostics became easier to trace and debug.
- Integrated **JUnit**, **Mockito**, and **Cucumber** tests into **Jenkins** CI/CD pipelines with branch-aware triggers and nightly regression runs and integrated **SonarQube** for code quality and coverage validation.
- Participated in **Agile** ceremonies to contribute backend testing insights, prioritize defect triaging, and estimate task sizing for each sprint.
- Set up **Jenkins** pipelines to deploy backend test services from **Github repository** into **ECS** using **blue/green** deployment strategies so that test coverage was not impacted during rollouts.

Environment: Java 17/21, Angular 16/19, Spring Boot, Python, Spring WebFlux, Microservices, TypeScript, JUnit, Mockito, TestNG, Cucumber, OpenTelemetry, Prometheus, RESTful APIs, API gateway, AWS ECS, AWS CloudWatch, AWS IAM, AWS S3, Jenkins, Docker, Kafka, OAuth2, MySQL, MongoDB, NoSQL, Grafana, DynamoDB, Splunk, SonarQube, Github, Postman, Swagger, Lambda, ConcurrentHashMap, JSON, Git, JIRA, CI/CD, Agile/Scrum, AI, Github Copilot

Client: JP Morgan Chase, Wilmington, DE
Software Engineer

Aug 2020 to Jun 2022

Project: Equity Insights Dashboard

Developed a web-based platform to display aggregated equity data, historical trends, and symbol **metadata** for internal analysts and product teams. Backend services built with **Spring Cloud** provided **REST** endpoints to serve processed data from **PostgreSQL**, while **Angular** components enabled filtering by ticker, sector, and market cap.

Responsibilities:

- Designed and developed **Java 11** Web and batch applications using **Spring Boot** and microservice frameworks such as **Spring Cloud Netflix Eureka**, **Config Server**, and **Gateway**.
- Created responsive UI components using **Angular 14**, **TypeScript**, **HTML5**, **Bootstrap**, and **JQuery** for enterprise web applications.
- Built backend modules with **Hibernate** and **JPA** for data persistence and implemented **transactional logic** across service layers.
- Developed and maintained **Spring Boot RESTful APIs** in a **microservices-based architecture** integrated with **AWS** RDS and **PostgreSQL**.

- Added detailed **Swagger/OpenAPI** documentation for REST endpoints to standardize API specifications and streamline integration between backend and frontend teams.
- Implemented **Infrastructure-as-Code** using **Terraform** to provision resources, integrated with Jenkins **CI/CD** pipelines for automated environment creation and deployments.
- Performed **state management** using **RxJS** and integrated **NgRx** in complex Angular modules to maintain consistent application state.
- Deployed Spring Boot microservices on **AWS ECS** and **Kubernetes** clusters using Jenkins pipelines to achieve scalable, containerized deployments
- Designed optimized **PostgreSQL schemas**, refactored SQL queries for **Oracle-to-PostgreSQL migration**, and integrated **Flyway** for version-controlled schema updates.
- Integrated **MongoDB** as a NoSQL layer for caching and hybrid data storage with **PostgreSQL** and **DynamoDB**.
- Performed **unit** and **integration testing** using **JUnit** and **Mockito** for services, web, and batch flows.
- Developed and integrated **Kafka producers** and **consumers** in Spring Boot microservices for asynchronous data processing and real-time equity insights.
- Created **Kafka topics** and **consumer groups** to handle real-time data streams efficiently.
- Configured **AWS API Gateway** to route secure traffic to microservices with **JWT-based authentication** and throttling policies.
- Implemented **OAuth2** and **JWT-based authentication** with **RBAC policies** to secure internal API endpoints and user roles.
- Integrated **Angular Material** and third-party libraries to maintain a consistent and responsive UI design.
- Investigated **production** and **SIT issues** by analyzing logs in **Splunk** and deployed necessary backend fixes.
- Containerized Spring Boot applications using **Docker** and deployed them to **Kubernetes** clusters for uniform, portable build environments.
- Supported **Independent Verification and Validation (IV&V)** efforts by collaborating with QA teams and addressing system test observations.
- Configured **Kubernetes** deployments, services, and ingress controllers to manage container orchestration, scaling, and load balancing for microservices.
- Collaborated with cross-functional teams on **system architecture**, interface design, and **performance tuning** based on business requirements.
- Integrated **Jenkins CI/CD pipelines** with **SonarQube** for static code analysis, automated testing, and **zero-downtime deployments**.
- Assisted with **software deployments** and used **AWS CloudWatch** and **Splunk** for monitoring resource usage and service health.
- Used **Git** and **Bitbucket** for version control and managed automated build and deployment pipelines via **Jenkins**.

Environment: Java 8/11, Angular 14, Spring Boot, Spring Cloud (Eureka, Config Server, Gateway), Microservices, TypeScript, RxJS, NgRx, HTML5, CSS3, Bootstrap, JQuery, Swagger/OpenAPI, Hibernate, JPA, Kafka, OAuth2, JWT, RBAC, AWS (ECS, EKS, EC2, RDS, S3, IAM, CloudWatch), Terraform, PostgreSQL, Flyway, Oracle, MongoDB, DynamoDB, Git, Kubernetes, Docker, Jenkins, SonarQube, JUnit, Mockito, Git, Bitbucket, Splunk, Agile/Scrum.

Client: Nike, Beaverton, OR
Role: Junior Java Developer

April 2019 to July 2020

Project: Inventory Visibility Tracker

Developed a web application to track product inventory levels across warehouses and retail locations, helping operations teams respond to stock fluctuations. Used **Spring MVC** and **JPA** to build backend services that exposed inventory data through REST endpoints, while **Angular** components displayed real-time stock levels, low-inventory alerts, and restock status.

Responsibilities:

- Developed and maintained enterprise-level web applications using **Java**, **J2EE**, **JSP**, **Servlets**, and **Spring** frameworks.
- Implemented **feature flags** in Angular modules to allow **QA** users to toggle test logic and UI features without requiring code deployments.
- Built dynamic, server-driven web pages using **JSP**, and implemented backend logic with **Servlets** and **EJBs**.
- Configured and implemented **Angular** routing, including route guards, to manage navigation and secure access to specific parts of the application.
- Wrote business logic and data access layers using **Hibernate** integrated with **JPA** to perform **Object Relational Mapping (ORM)** between Java classes and Oracle database tables.
- Developed and consumed **SOAP** and **REST** based web services to support communication between internal and external systems.
- Integrated **Angular Material** and other third-party libraries to maintain a consistent and responsive UI aligned with modern design standards.
- Applied **Spring MVC** architecture to implement presentation logic, controllers, and model components using **Java Beans**.
- Used **Dependency Injection (DI)** annotations and XML-based configuration for bean management and decoupling components within the **Spring IOC** container.
- Wrote and executed SQL queries, PL/SQL procedures, and functions for data retrieval, reporting, and transactional support in **Oracle**.
- Implemented **MVC** design pattern to separate concerns and applied **J2EE** best practices including **DAO**, **Singleton**, and **Factory** patterns for reusable and maintainable code.
- Created **XSLT** transformations for converting **XML** documents into **HTML** output to improve data readability for web users.
- Developed client-side functionality using **HTML**, **CSS**, **JavaScript**, **jQuery**, and **AJAX** to provide interactive user experiences.
- Configured and deployed web applications on **Apache Tomcat** and participated in integration activities with other application modules.
- Used **SVN** for version control and collaborated with team members on source code updates and merge activities.
- Automated build and deployment processes using **Maven** and integrated build triggers with **Jenkins** for Continuous Integration.

Environment: Java 1.8, Angular 10, J2EE, JSP, Servlets, EJB, Spring, Spring MVC, Spring Batch, Spring IOC, Hibernate, JPA, SOAP, REST, SQL, PL/SQL, Oracle, HTML, CSS, JavaScript, jQuery, AJAX, JSON, XML, XSLT, Maven, Jenkins, SVN, Apache Tomcat